

```
In [1]: import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [2]: income_df = pd.read_csv(r'/Users/abhiramireddygal/Downloads/april(
```

```
In [3]: income_df.head()
```

```
Out[3]:
```

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_
0	5000	8000	3	
1	6000	7000	2	
2	10000	4500	2	
3	10000	2000	1	
4	12500	12000	2	

```
In [4]: income_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Mthly_HH_Income                       50 non-null     int64
1   Mthly_HH_Expense                       50 non-null     int64
2   No_of_Fly_Members                     50 non-null     int64
3   Emi_or_Rent_Amt                       50 non-null     int64
4   Annual_HH_Income                      50 non-null     int64
5   Highest_Qualified_Member              50 non-null     object
6   No_of_Earning_Members                 50 non-null     int64
dtypes: int64(6), object(1)
memory usage: 2.9+ KB
```

```
In [5]: income_df.shape
```

```
Out[5]: (50, 7)
```

```
In [7]: income_df.describe().T
```

```
Out[7]:
```

	count	mean	std	min	25%
Mthly_HH_Income	50.0	41558.00	26097.908979	5000.0	23550.0
Mthly_HH_Expense	50.0	18818.00	12090.216824	2000.0	10000.0
No_of_Fly_Members	50.0	4.06	1.517382	1.0	3.0
Emi_or_Rent_Amt	50.0	3060.00	6241.434948	0.0	0.0
Annual_HH_Income	50.0	490019.04	320135.792123	64200.0	258750.0
No_of_Earning_Members	50.0	1.46	0.734291	1.0	1.0

```
In [8]: income_df.isna().any()
```

```
Out[8]: Mthly_HH_Income      False
Mthly_HH_Expense      False
No_of_Fly_Members      False
Emi_or_Rent_Amt        False
Annual_HH_Income       False
Highest_Qualified_Member False
No_of_Earning_Members  False
dtype: bool
```

```
In [9]: income_df['Mthly_HH_Expense'].mean()
```

```
Out[9]: np.float64(18818.0)
```

```
In [14]: income_df['Mthly_HH_Expense'].mode()
```

```
Out[14]: 0    25000
Name: Mthly_HH_Expense, dtype: int64
```

```
In [10]: income_df['Mthly_HH_Expense'].median()
```

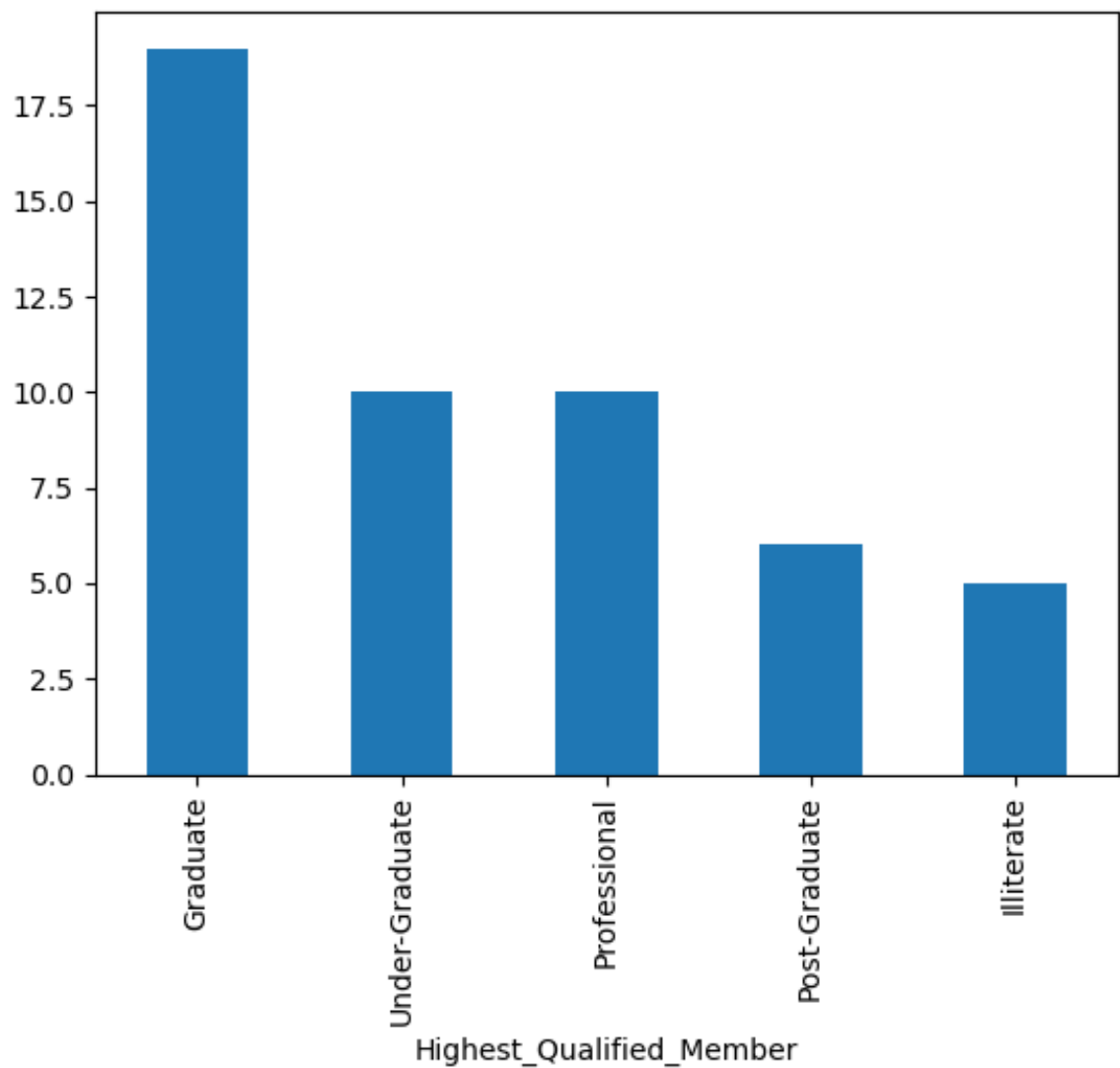
```
Out[10]: 15500.0
```

```
In [13]: mth_exp_tmp = pd.crosstab(index=income_df['Mthly_HH_Expense'], columns=income_df['Highest_Qualified_Member'],
mth_exp_tmp.reset_index(inplace = True)
mth_exp_tmp[mth_exp_tmp['count']] == income_df.Mthly_HH_Expense.value_counts()
```

```
Out[13]: col_0  Mthly_HH_Expense  count
18          25000          8
```

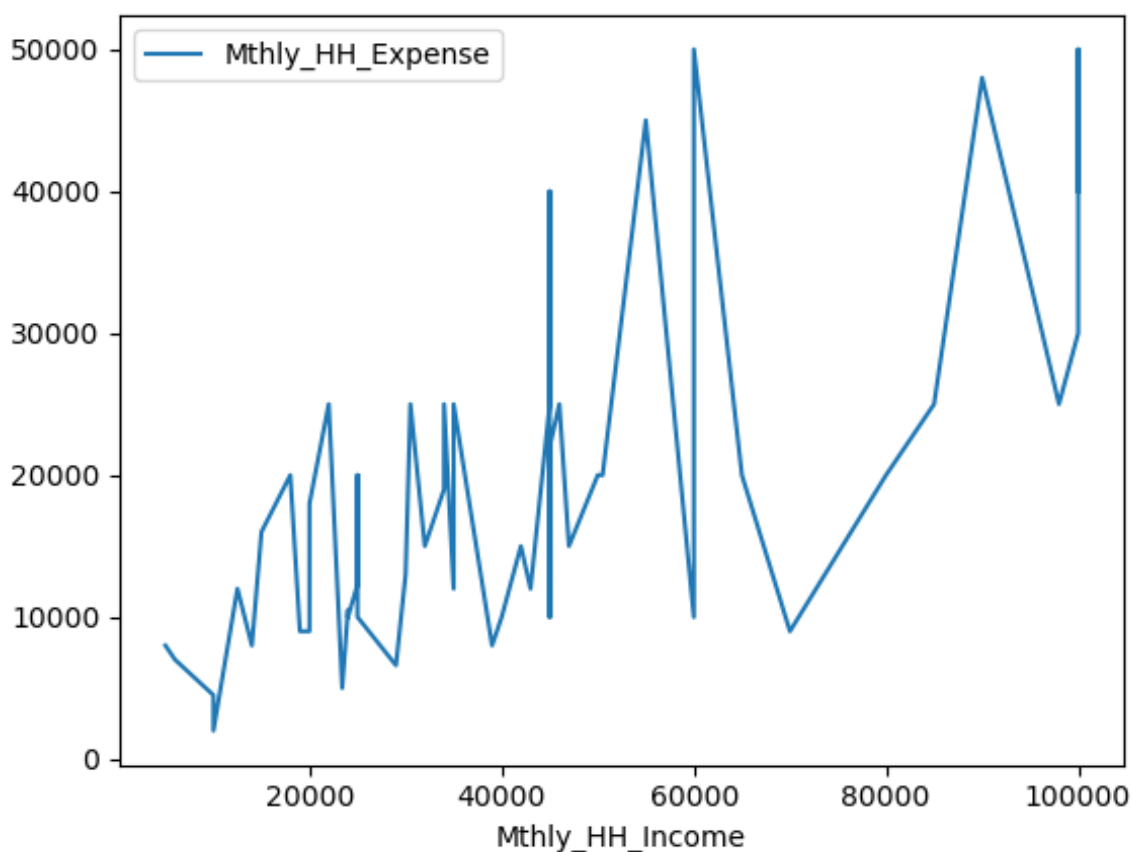
```
In [15]: income_df['Highest_Qualified_Member'].value_counts().plot(kind = 'bar')
```

```
Out[15]: <Axes: xlabel='Highest_Qualified_Member'>
```



```
In [17]: income_df.plot(x = 'Mthly_HH_Income', y = 'Mthly_HH_Expense')
IQR = income_df['Mthly_HH_Expense'].quantile(0.75)-income_df['Mthly_
IQR
```

```
Out[17]: np.float64(1450.0)
```



```
In [18]: pd.DataFrame(income_df.iloc[:,0:5].std().to_frame()).T
```

```
Out[18]:
```

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_
0	26097.908979	12090.216824	1.517382	6241.43

```
In [19]: income_df["Highest_Qualified_Member"].value_counts().to_frame().T
```

```
Out[19]:
```

Highest_Qualified_Member	Graduate	Under-Graduate	Professional	Post-Graduate	Illite
count	19	10	10	6	

```
In [20]: income_df["No_of_Earning_Members"].value_counts().plot(kind="bar")
```

```
Out[20]: <Axes: xlabel='No_of_Earning_Members'>
```

